Roll No. .....

TBT–402

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

IMMUNOLOGY AND IMMUNOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) With the help of diagram only, explain different types of immune cells. (CO2)

OR

(b) Comment briefly on contribution of various scientists in the field of Immunology. (CO1)

2. (a) What is phagocytosis ? Explain the mechanism of killing through phagocytosis. (CO1)

OR

(b) What are the pharmacological indicators of inflammation ? Explain their significance in inflammation process. (CO1)

3. (a) Using suitable diagram, explain the structure and function of thymus. (CO2)

P. T. O.

OR

(b) Explain the structure of immunoglobulin. What are the different isotypes of immunoglobulins based on their variation in constant region ? (CO2)

4. (a) Differentiate between humoral and cell mediated immune response. (CO2)

OR

(b) Explain the significance of anatomical and physiological barriers of innate immunity. (CO2)

5. (a) What are complement proteins ? Explain the principle of complement fixation test. (CO3)

OR

(b) What are antigen presenting cells ? Explain their role in T-cell activation. (CO2)

OR

(b) Explain the structure of immunoglobulin. What are the different isotypes of immunoglobulins based on their variation in constant region ? (CO2)

4. (a) Differentiate between humoral and cell mediated immune response. (CO2)

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(b) Explain the significance of anatomical and physiological barriers of innate immunity. (CO2)

5. (a) What are complement proteins ? Explain the principle of complement fixation test. (CO3)

OR

(b) What are antigen presenting cells ? Explain their role in T-cell activation. (CO2)

Roll No. .....

TBT–405(a)

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

WASTE MANGEMENT AND UPCYCLING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *one* question among (a) and (b) in each main question.

(iii) Total marks in each main question are ten.

(iv) Each question carries 10 marks.

1. (a) What is the importance of waste management ? (CO1/CO2)

OR

(b) Define waste and write down its classification. (CO1/CO2)

2. (a) Explain operation of waste stabilization pond system for waste water treatment. (CO1C/CO2)

OR

(b) What are the measures for managing waste ? How does waste upcycling differ from waste recycling ? (CO1/CO2)

3. (a) How does Reed beds help in effluent treatment ? Explain with the help of a suitable diagram. (CO1/CO2)

P. T. O.

OR

(b) Mention the principle and functions of any *two* Air pollution control devices. (CO1/CO2)

4. (a) What are greenhouse gases ? How does carbon sequestration help in removal of CO_2 from the environment ? (CO1/CO2)

OR

(b) Give a detailed account of sources of industrial wastewater and their environmental impacts. (CO1/CO2)

5. (a) Explain how Scrubbers help in removal of toxic materials from flue gas. (CO2)

OR

(b) Explain carbon capture utilization and storage and mention its benefits. (CO2)

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TBT–405(C)

B. TECH. (BIOTECH.) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

BIOMATERIALS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the evolution of biomateria from 1st generation to 4th generation. (CO1/CO1)

OR

(b) Write the chemical bonds present in the metal, polymer, and ceramics and their effects on properties. (CO1/CO1)

2. (a) What is the significance of composite in biomaterial ? Explain with examples. (CO1/CO1)

OR

(b) Define the following on the basis of constituents, mechanical and tribological properties : (CO1/CO1)

(i) Mineral fiber

(ii) Animal fiber

P. T. O.

3. (a) Write short notes on the following : (CO1/CO1)

(i) Fracture toughness

(ii) Hardness

(iii) Ductility

(iv) Refractoriness

(v) Compressive strength

OR

(b) What are bioceramics ? Discuss any *two* medical applications where ceramics-based biomaterials are used. (CO1/CO1)

4. (a) Discuss the nucleation and growth mechanism. How does the rate of cooling affect the grain and grain boundary formation ? Explain in brief. (CO1/CO2)

OR

(b) What would be the greatest length of a steel wire which when fixed at one end can hang freely without breaking ? Density of steel = 7800 kg/m^3. Breaking stress for steel = 7.8×10^8 N/m^2. (CO1/CO2)

5. (a) A wire 2 m long and 2 mm in diameter, when stretched by weight of 8 kg has its length increased by 0.24 mm. Find the stress, strain and Young's modulus of the material of the wire. $g = 9.8$ m/s^2. (CO2/CO2)

OR

(b) What must be the elongation of a wire 5m long so that the strain is 1% of 0.1 ? If the wire has cross-selection of 1 mm^2 and is stretched by 10 kg-wt, what is the stress ? (CO2/CO2)

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TBT–403

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

BIOANALYTICAL TECHNIQUES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Differentiate between SEM and TEM. (CO1)

OR

(b) Differentiate between Bright field and Dark field microscopy. (CO1)

2. (a) Explain the working of fluorescence microscopy and its importance. (CO1)

OR

(b) Discuss confocal microscopy and its application. (CO1)

3. (a) Explain the term bioanalytical technique and its importance in biotechnology field. (CO1/CO2)

OR

(b) Explain various factors that affect centrifugations process with the help of equation. (CO1/CO2)

P. T. O.

4. (a) Differentiate between differential and density gradient centrifugation. (CO2)

OR

(b) Write short notes on the following : (CO2)

(i) Role of sucrose and glycerol in 2D SDS-PAGE

(ii) Role of ampholyte molecule in 2D SDS-PAGE

(iii) Role of SDS and PAGE in electrophoresis

5. (a) Draw a relationship between different forces acting during centrifugation process. (CO2)

OR

(b) Explain the term immunoelectrophoresis. (CO2)

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TBT–404

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

ENVIRONMENTAL BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the relevance of the biotechnological approaches that one should apply for the protection of the environment. (CO1/CO2)

OR

(b) Describe the methods adopted for the measurement of pollution. (CO1/CO2)

2. (a) "A form of precipitation which has high level of hydrogen ions." Explain such form and its impact with the help of diagram. (CO1)

OR

(b) Highlight the current status and the problem solving approaches for the ozone layer depletion. (CO1)

3. (a) How will you define sludge ? Explain the treatment methods involved for it. (CO2)

P. T. O.

OR

(b) Describe the role and mechanism of thermal hydrolysis plant with the help of a diagram. (CO2)

4. (a) Differentiate between Chemical Oxygen demand and Biochemical Oxygen Demand. (CO1/CO2)

OR

(b) "Process uses microorganisms to feed on organic contaminants in wastewater, producing a purified effluent." Explain such process diagrammatically. (CO1/CO2)

5. (a) Explain the steps involved in the construction of sanitary landfill. (CO2)

OR

(b) Describe the management strategies that should be adopted for solid waste and the processing of organic residues. (CO2)

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TBT–401

B. TECH. (BIOTECH.) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

BIOPHYSICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the molecular orbital theory using the example of H_2 molecule. (CO1)

OR

(b) Write a short note on hybridization of atomic orbitals. (CO1)

2. (a) Briefly describe the principles of XRD and its applications in life science. (CO1/CO2)

OR

(b) Explain the mechanism of circular dichroism (CD) spectroscopy. (CO1/CO2)

3. (a) Describe the role of enzymes in biological process. (CO1)

OR

(b) Explain the Lennard-Jones potential for weak interactions. (CO1)

P. T. O.

4. (a) Explain the hierarchical structure of protein. (CO2)

OR

(b) Describe the Ramchandran plot for protein structure. (CO2)

5. (a) Write a brief essay on the thermodynamics of protein folding. (CO2)

OR

(b) Explain the first, second and third laws of thermodynamics. (CO2)

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3. (a) A fair die is tossed twice. Find the probability of getting a 4, 5, or 6 on the first toss and a 1, 2, 3, or 4 on the second toss. (CO3)

OR

(b) Ram and Shyam play 12 games of chess, of which 6 are won by A, 4 are won by B, and 2 end in a draw. They agree to play a match consisting of 3 games. Find the probability that (i) A wins all 3 games, (ii) 2 games end in a draw, (iii) A and B win alternately and (iv) B wins at least 1 game. (CO3)

4. (a) How is standard packages different from command packages in R ? Illustrate the steps for Installing Extra Packages for Windows Users. (CO2)

OR

(b) Explain the motivations behind using R as compared to Python. (CO2)

5. (a) Create two objects data1 and data2 of weekdays and months in a year respectively. Write a function in R to save these objects with proper extension, remove these objects, check these objects and recovery of these objects in workspace. (CO3)

OR

(b) You have a dataset named test.csv having multiple entries with two columns the data is to be read along with the header, in the second scenario there is delimiter space in same file. How to read these files and handle missing values in R ? (CO3)